

Fundamentals of Information Systems Security

Chapter 5 Quiz: Access Controls

Name: _____

Date: _____

Total Points: 100

Time Allowed: 30 Minutes

Instructions

- This quiz contains **20 questions** covering Chapter 5
- Answer all questions to the best of your ability
- Multiple choice questions have only one correct answer
- True/False questions should be marked T or F
- Short answer questions require concise, accurate responses

Section A: Multiple Choice (40 points)

1. (4 points) What are the four parts of access control in order?
 - A. Identification, Authorization, Authentication, Accountability
 - B. Identification, Authentication, Authorization, Accountability
 - C. Authentication, Identification, Authorization, Accountability
 - D. Authorization, Authentication, Identification, Accountability
2. (4 points) Which type of access control allows the resource owner to decide who gets access?
 - A. Mandatory Access Control (MAC)
 - B. Discretionary Access Control (DAC)
 - C. Role-Based Access Control (RBAC)
 - D. Rule-Based Access Control
3. (4 points) What is the crossover error rate (CER) in biometrics?
 - A. The point where false rejection rate equals false acceptance rate
 - B. The point where authentication fails completely
 - C. The rate of system failure
 - D. The rate of user rejection
4. (4 points) Which authentication factor is classified as "something you have"?
 - A. Password
 - B. Fingerprint
 - C. Smart card

- D. PIN
5. (4 points) What is the principle of least privilege?
- A. Giving users maximum access
 - B. Giving users the minimum access necessary to perform their job
 - C. Giving users access based on seniority
 - D. Giving users access to all resources
6. (4 points) Which access control model uses security labels and clearances?
- A. Discretionary Access Control (DAC)
 - B. Mandatory Access Control (MAC)
 - C. Role-Based Access Control (RBAC)
 - D. Content-Dependent Access Control
7. (4 points) What does MFA stand for?
- A. Multi-Factor Authentication
 - B. Multi-Function Access
 - C. Mandatory Function Authentication
 - D. Major Factor Authorization
8. (4 points) Which AAA protocol is the most popular for network access authentication?
- A. TACACS+
 - B. DIAMETER
 - C. RADIUS
 - D. SAML
9. (4 points) What is the security kernel?
- A. The central part of a computing environment that enforces access control
 - B. A type of malware
 - C. A network security device
 - D. An encryption algorithm
10. (4 points) What is the difference between identification and authentication?
- A. Identification is proving identity; authentication is claiming identity
 - B. Identification is claiming identity; authentication is proving identity
 - C. They are the same thing
 - D. Identification is for systems; authentication is for users

Section B: True/False (20 points)

11. (2 points) Physical access controls control access to computer systems and networks.
- A. True
 - B. False
12. (2 points) Biometrics can be used for both identification and authentication.

- A. True
 - B. False
13. (2 points) Single Sign-On (SSO) allows users to log in once and access multiple systems.
- A. True
 - B. False
14. (2 points) Kerberos is a protocol that provides mutual authentication between user and server.
- A. True
 - B. False
15. (2 points) The Bell-LaPadula model focuses on integrity, not confidentiality.
- A. True
 - B. False
16. (2 points) Access control lists (ACLs) are used to define permissions on objects.
- A. True
 - B. False
17. (2 points) The Biba model focuses on confidentiality of data.
- A. True
 - B. False
18. (2 points) Cloud computing eliminates the need for access controls.
- A. True
 - B. False
19. (2 points) Accountability is the process of tracing actions to individuals.
- A. True
 - B. False
20. (2 points) Temporal isolation restricts access based on time of day.
- A. True
 - B. False

Section C: Short Answer (40 points)

21. (10 points) Explain the difference between DAC, MAC, and RBAC. Provide an example of when each would be used.

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22. (10 points) Describe the three types of authentication factors (something you know, have, are) and give two examples of each.
23. (10 points) What is the advantage and disadvantage of Single Sign-On (SSO)?
24. (10 points) Explain how Kerberos authentication works. What is the role of the Key Distribution Center (KDC)?

End of Quiz